

Epistasis between mutator alleles contributes to germline mutation rate variability in laboratory mice

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Abstract

Maintaining germline genome integrity is essential and enormously complex. Hundreds of proteins are involved in DNA replication and proofreading, and hundreds more are mobilized to repair DNA damage [1]. While loss-of-function mutations in any of the genes encoding these proteins might lead to elevated mutation rates, *mutator alleles* have largely eluded detection in mammals.

DNA replication and repair proteins often recognize particular sequence motifs or excise lesions at specific nucleotides. Thus, we might expect that the spectrum of *de novo* mutations — that is, the frequency of each individual mutation type (C>T, A>G, etc.) — will differ between genomes that harbor either a mutator or wild-type allele at a given locus. Previously, we used quantitative trait locus mapping to discover candidate mutator alleles in the DNA repair gene *Mutyh* that increased the C>A germline mutation rate in a family of inbred mice known as the BXDs [2, 3].

In this study we developed a new method, called “inter-haplotype distance,” to detect alleles associated with mutation spectrum variation. By applying this approach to mutation data from the BXDs, we confirmed the presence of the germline mutator locus near *Mutyh* and discovered an additional C>A mutator locus on chromosome 6 that overlaps *Ogg1* and *Mbd4*, two DNA glycosylases involved in base-excision repair [4, 5]. The effect of a chromosome 6 mutator allele depended on the presence of a mutator allele near *Mutyh*, and BXDs with mutator alleles at both loci had even greater numbers of C>A mutations than those with mutator alleles at either locus alone.

Our new methods for analyzing mutation spectra reveal evidence of epistasis between germline mutator alleles, and may be applicable to mutation data from humans and other model organisms.

eLife assessment

By developing a novel method for detecting genetic variants associated with germline mutation spectrum variation, this **important** study identifies a new "mutator" locus in a population of inbred mouse strains, although the causal gene(s) within this locus remains uncertain. The authors further demonstrate that this new mutator locus interacts epistatically with a previously identified mutator allele on C>A mutation rate, showcasing the complexity of the genetic basis underlying variation in mutation rate and spectrum. Evidence for major findings in this paper is **convincing**, but the power, statistical properties and limitations of the new method need to be further characterized to support its broader applicability to other systems.

Introduction

Germline mutation rates reflect the complex interplay between DNA proofreading and repair pathways, exogenous sources of DNA damage, and life-history traits. For example, parental age is an important determinant of mutation rate variability; in many mammalian species, the number of germline *de novo* mutations observed in offspring increases as a function of paternal and maternal age [6, 7, 8, 9, 10]. Rates of germline mutation accumulation are also variable across human families [7, 11], likely due to either genetic variation or differences in environmental exposures. Although numerous protein-coding genes contribute to the maintenance of genome integrity, genetic variants that increase germline mutation rates, known as *mutator alleles*, have proven difficult to discover in mammals.

The dearth of observed germline mutators in mammalian genomes is not necessarily surprising, since alleles that lead to elevated germline mutation rates would likely have deleterious consequences and be purged by negative selection [12]. Moreover, germline mutation rates are relatively low, and direct mutation rate measurements require whole-genome sequencing data from both parents and their offspring. As a result, large-scale association studies — which have been used to map the contributions of common genetic variants to many complex traits — are not currently well-powered to investigate the polygenic architecture of germline mutation rates [13].

Despite these challenges, less traditional strategies have been used to identify a small number of mutator alleles in humans, macaques, and mice [14]. By focusing on families with rare genetic diseases, a recent study discovered two mutator alleles that led to significantly elevated rates of *de novo* germline mutation in human genomes [15]. Another group observed mutator phenotypes in the sperm and somatic tissues of adults who carry cancer-predisposing inherited mutations in the POLE/POLD1 exonucleases [16]. Candidate mutator loci were also found by identifying human haplotypes from the Thousand Genomes Project with excess counts of derived alleles in genomic windows [17].

In mice, a germline mutator allele was recently discovered by sequencing a large family of inbred mice [2]. Commonly known as the BXDs, these recombinant inbred lines (RILs) were derived from either F2 or advanced intercrosses of C57BL/6J and DBA/2J, two laboratory strains that exhibit significant differences in their germline mutation spectra [3, 18]. The BXDs were maintained via brother-sister mating for up to 180 generations, and each BXD therefore accumulated hundreds or thousands of germline mutations on a nearly-homozygous linear

mosaic of parental *B* and *D* haplotypes. Due to their husbandry in a controlled laboratory setting, the BXDs were largely free from confounding by environmental heterogeneity, and the effects of selection on *de novo* mutations were attenuated by strict inbreeding [19].

In this previous study, whole-genome sequencing data from the BXD family were used to map a quantitative trait locus (QTL) for the C>A mutation rate [2]. Germline C>A mutation rates were nearly 50% higher in mice with *D* haplotypes at the QTL, likely due to genetic variation in the DNA glycosylase *MutYh* that reduced the efficacy of oxidative DNA damage repair. Importantly, the QTL did not reach genome-wide significance in a scan for variation in overall germline mutation rates, which were only modestly higher in BXDs with *D* alleles, demonstrating the utility of mutation spectrum analysis for mutator allele discovery. Close examination of the mutation spectrum is likely to be broadly useful for detecting mutator alleles, as genes involved in DNA proofreading and repair often recognize particular sequence motifs or excise specific types of DNA lesions [20]. Mutation spectra are usually defined in terms of *k*-mer nucleotide context; the 1-mer mutation spectrum, for example, consists of 6 mutation types after collapsing by strand complement (C>T, C>A, C>G, A>T, A>C, A>G), while the 3-mer mutation spectrum contains 96 (each of the 1-mer mutations partitioned by trinucleotide context).

Although mutation spectrum analysis can enable the discovery of mutator alleles that affect the rates of specific mutation types, early implementations of this strategy have suffered from a few drawbacks. For example, performing association tests on the rates or fractions of every *k*-mer mutation type can quickly incur a substantial multiple testing burden. Since germline mutation rates are generally quite low, estimates of *k*-mer mutation type frequencies from individual samples can also be noisy and imprecise. We were therefore motivated to develop a statistical method that could overcome the sparsity of *de novo* mutation spectra, eliminate the need to test each *k*-mer mutation type separately, and enable sensitive detection of alleles that influence the germline mutation spectrum.

Here, we present a new mutation spectrum association test, called “inter-haplotype distance,” that minimizes multiple testing burdens and mitigates the challenges of sparsity in *de novo* mutation datasets. We leverage this method to re-analyze germline mutation data from the BXD family and find compelling evidence for a second mutator allele that was not detected using previous approaches. The new allele appears to interact epistatically with the mutator that was previously discovered in the BXDs, further augmenting the C>A germline mutation rate in a subset of inbred mice. Our observation of epistasis suggests that mild DNA repair deficiencies can compound one another, as mutator alleles chip away at the redundant systems that collectively maintain germline integrity.

Results

A novel method for detecting mutator alleles

We developed a statistical method, termed “inter-haplotype distance” (IHD), to detect loci that are associated with mutation spectrum variation in recombinant inbred lines (RILs) (Figure 1; *Materials and Methods*). Our approach leverages the fact that mutator alleles often leave behind distinct and detectable impressions on the *mutation spectrum*, even if they increase the overall mutation rate by a relatively small amount. Given a population of haplotypes, we assume that each has been genotyped at the same collection of biallelic loci and that each harbors *de novo* mutations which have been partitioned by *k*-mer context (Figure 1). At every locus, we calculate a cosine distance between the aggregate mutation spectra of haplotypes that inherited either parental allele. Using permutation tests, we then identify loci at which those distances are larger than what we’d expect by random chance. To account for polygenic effects on the mutation

process that might be shared between BXDs, we also regress the cosine distance at each marker against the genetic similarity between haplotype groups, and assess significance using the fitted residuals (which we call the “adjusted” cosine distances) (*Materials and Methods*).

Using simulated data, we find that our method’s power is primarily limited by the initial mutation rate of the *k*-mer mutation type affected by a mutator allele and the total number of *de novo* mutations used to detect it (**Figure 1** [figure supplement 1](#)). Given 100 haplotypes with an average of 500 *de novo* germline mutations each, IHD has approximately 90% power to detect a mutator allele that increases the C>A *de novo* mutation rate by as little as 20%. However, the approach has less than 20% power to detect a mutator of identical effect size that augments the C>G mutation rate, since C>G mutations are expected to make up a smaller fraction of all *de novo* germline mutations to begin with. Simulations also demonstrate that our approach is well-powered to detect large-effect mutator alleles (e.g., those that increase the mutation rate of a specific *k*-mer by 50%), even with a relatively small number of mutations per haplotype (**Figure 1** [figure supplement 1](#)). Both IHD and traditional quantitative trait locus (QTL) mapping have similar power to detect alleles that augment the rates of individual 1-mer mutation types (**Figure 1** [figure supplement 2](#)), but IHD has a number of potential advantages for mutator allele discovery; for a more detailed comparison of the methods, see the *Discussion*.

Re-identifying a mutator allele on chromosome 4 in the BXDs

We applied our inter-haplotype distance method to 117 BXDs (*Materials and Methods*) with a total of 65,552 *de novo* germline mutations [[2](#)]. Using mutation data that were partitioned by 1-mer nucleotide context, we discovered a locus on chromosome 4 that was significantly associated with mutation spectrum variation (**Figure 2a** [figure supplement 1](#); maximum adjusted cosine distance of 1.20e-2 at marker ID rs27509845; position 118.28 Mbp in GRCm38/mm10 coordinates).

Using quantitative trait locus (QTL) mapping, we previously identified a nearly-identical locus on chromosome 4 that was significantly associated with the C>A germline mutation rate in the BXDs [[2](#)]. This locus overlapped 21 protein-coding genes that were annotated by the Gene Ontology as being involved in “DNA repair,” but only one of those genes contained nonsynonymous differences between the two parental strains: *Mutyh*. *Mutyh* encodes a protein involved in the base-excision repair of 8-oxoguanine (8-oxoG), a DNA lesion caused by oxidative damage, and prevents the accumulation of C>A mutations [[4](#),[21](#),[22](#)]. C>A germline mutation fractions are nearly 50% higher in BXDs that inherit *D* genotypes at marker ID rs27509845 (the marker at which we observed the highest adjusted cosine distance on chromosome 4) than in those that inherit *B* genotypes (**Figure 3** [figure supplement 1](#)) [[2](#)].

An additional germline mutator allele on chromosome 6

After confirming that IHD could recover the mutator locus overlapping *Mutyh*, we asked if our approach could identify additional mutator loci in the BXDs. In particular, we were interested in discovering epistatic interactions between alleles at the chromosome 4 locus and mutator alleles elsewhere in the genome. We hypothesized that such interactions could be detectable by first “conditioning” on the presence of *B* or *D* alleles at the mutator locus on chromosome 4, and then running another genome-wide scan for loci associated with mutation spectrum variation. To account for the effects of the large-effect mutator locus near *Mutyh*, we divided the BXDs into those with either *D* (*n* = 66) or *B* (*n* = 44) genotypes at rs27509845 (*n* = 7 BXDs were heterozygous) and ran an inter-haplotype distance scan using each group separately (**Figure 2b-c** [figure supplement 1](#)).

Using the BXDs with *D* genotypes at rs27509845, we identified a locus on chromosome 6 that was significantly associated with mutation spectrum variation (**Figure 2b** [figure supplement 1](#); maximum adjusted cosine distance of 3.68e-3 at marker rs46276051; position 111.27 Mbp in GRCm38/mm10 coordinates). This signal was specific to BXDs with *D* genotypes at the rs27509845 locus, as we did

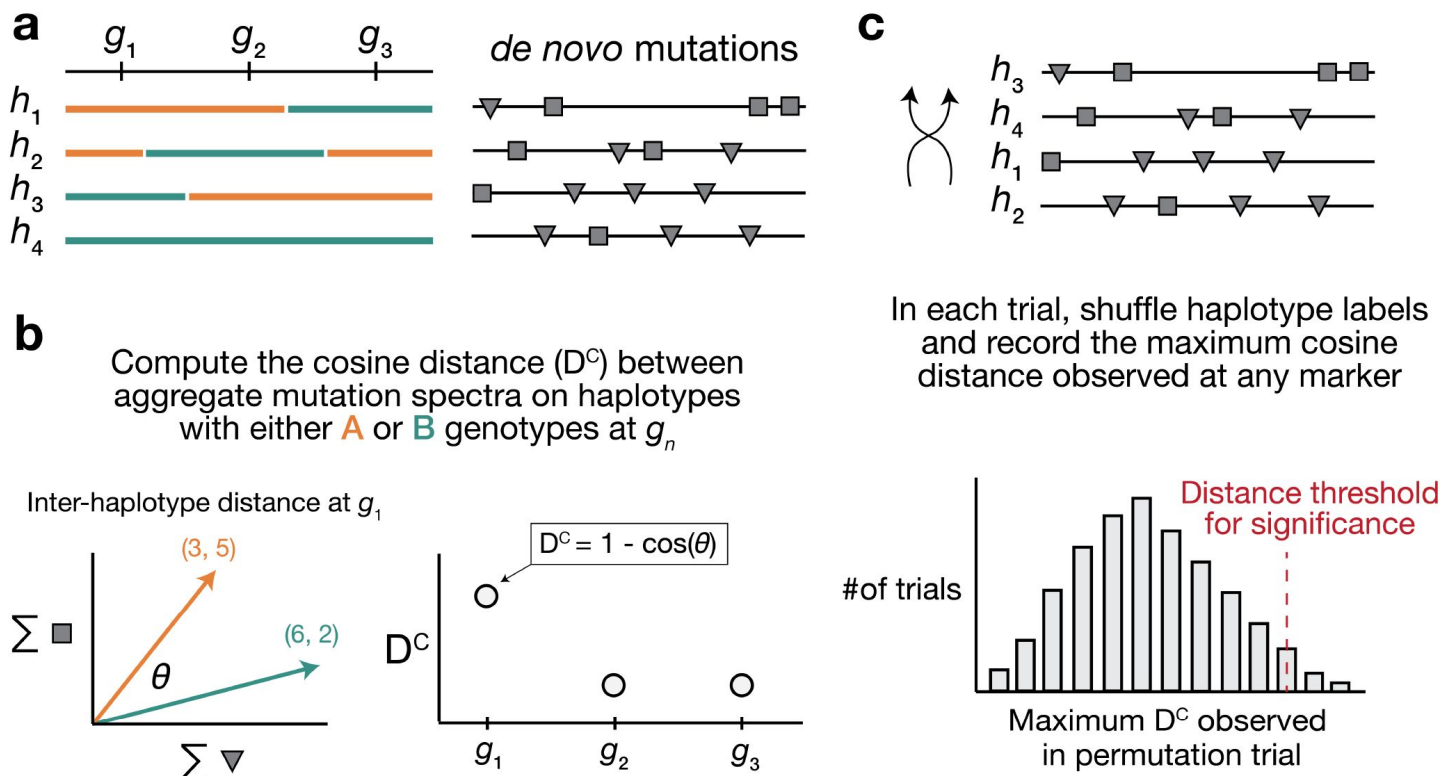


Figure 1

Overview of inter-haplotype distance method for discovering mutator alleles.

a) A population of four haplotypes has been genotyped at three informative markers (g_1 through g_3); each haplotype also harbors unique *de novo* germline mutations. In practice, *de novo* mutations are partitioned by k -mer context; for simplicity in this toy example, *de novo* mutations are simply classified into two possible mutation types (grey squares represent C>(A/T/G) mutations, while grey triangles represent A>(C/T/G) mutations). **b**) At each informative marker g_n , we calculate the total number of each mutation type observed on haplotypes that carry either parental allele (i.e., the aggregate mutation spectrum) using all genome-wide *de novo* mutations. For example, haplotypes with A (orange) genotypes at g_1 carry a total of three “triangle” mutations and five “square” mutations, and haplotypes with B (green) genotypes carry a total of six triangle and two square mutations. We then calculate the cosine distance between the two aggregate mutation spectra, which we call the “inter-haplotype distance.” Cosine distance can be defined as $1 - \cos(\theta)$, where θ is the angle between two vectors; in this case, the two vectors are the two aggregate spectra. We repeat this process for every informative marker g_n . **c**) To assess the significance of any distance peaks in b), we perform permutation tests. In each of N permutations, we shuffle the haplotype labels associated with the *de novo* mutation data, run a genome-wide distance scan, and record the maximum cosine distance encountered at any locus in the scan. Finally, we calculate the 1- p percentile of the distribution of those maximum distances to obtain a genome-wide cosine distance threshold at the specified value of p .

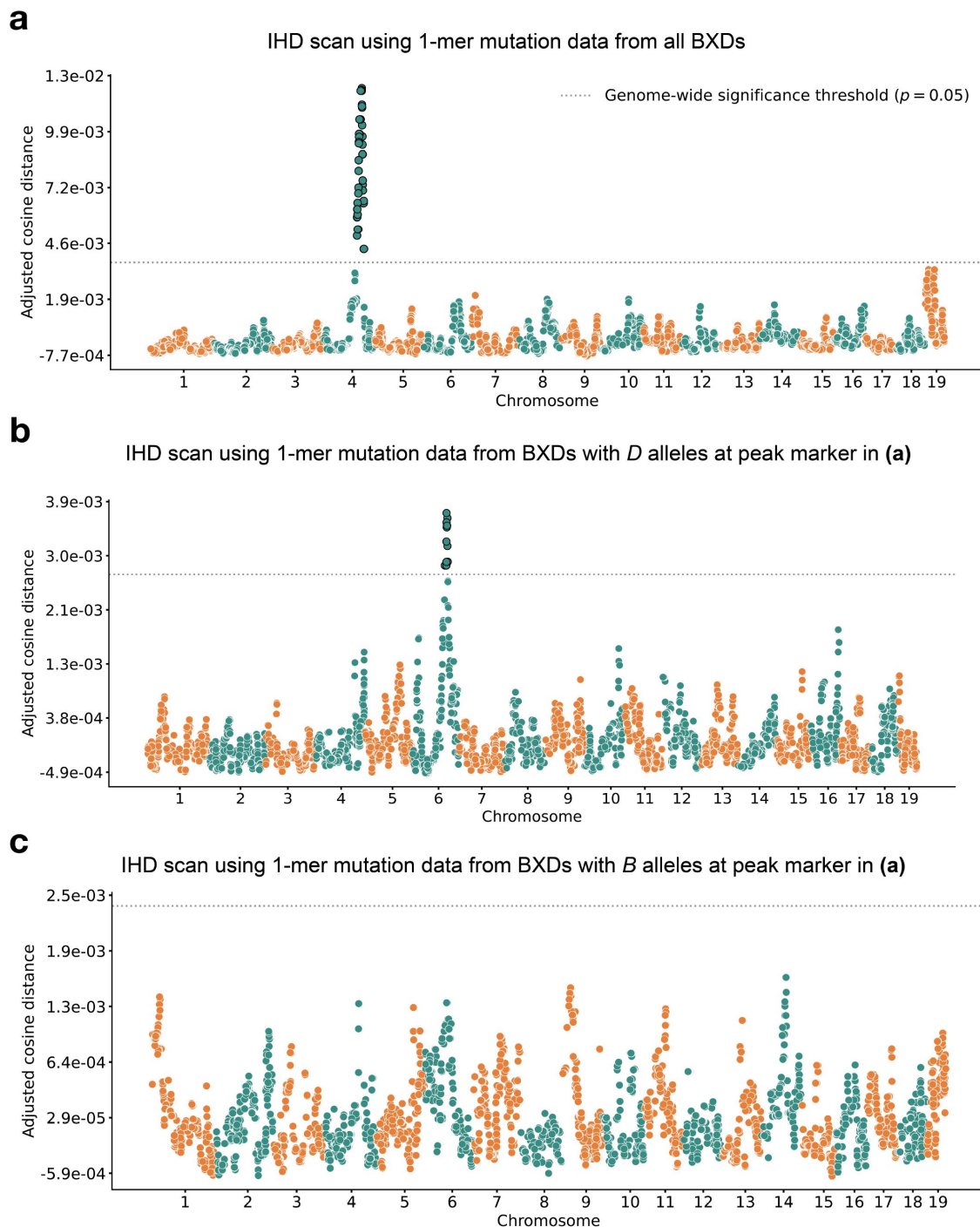


Figure 2

Results of inter-haplotype distance scans in the BXDs.

a) Adjusted cosine distances between aggregate 1-mer *de novo* mutation spectra on BXD haplotypes ($n = 117$ haplotypes; 65,552 total mutations) with either *D* or *B* alleles at 7,321 informative markers. Cosine distance threshold at $p = 0.05$ was calculated by performing 10,000 permutations of the BXD mutation data, and is shown as a dotted grey line. **b)** Adjusted cosine distances between aggregate 1-mer *de novo* mutation spectra on BXD haplotypes with *D* alleles at rs27509845 ($n = 66$ haplotypes; 42,171 total mutations) and either *D* or *B* alleles at 7,276 informative markers. Cosine distance threshold at $p = 0.05$ was calculated by performing 10,000 permutations of the BXD mutation data, and is shown as a dotted grey line. **c)** Adjusted cosine distances between aggregate 1-mer *de novo* mutation spectra on BXD haplotypes with *B* alleles at rs27509845 ($n = 44$ haplotypes; 22,645 total mutations) and either *D* or *B* alleles at 7,273 informative markers. Cosine distance threshold at $p = 0.05$ was calculated by performing 10,000 permutations of the BXD mutation data, and is shown as a dotted grey line.

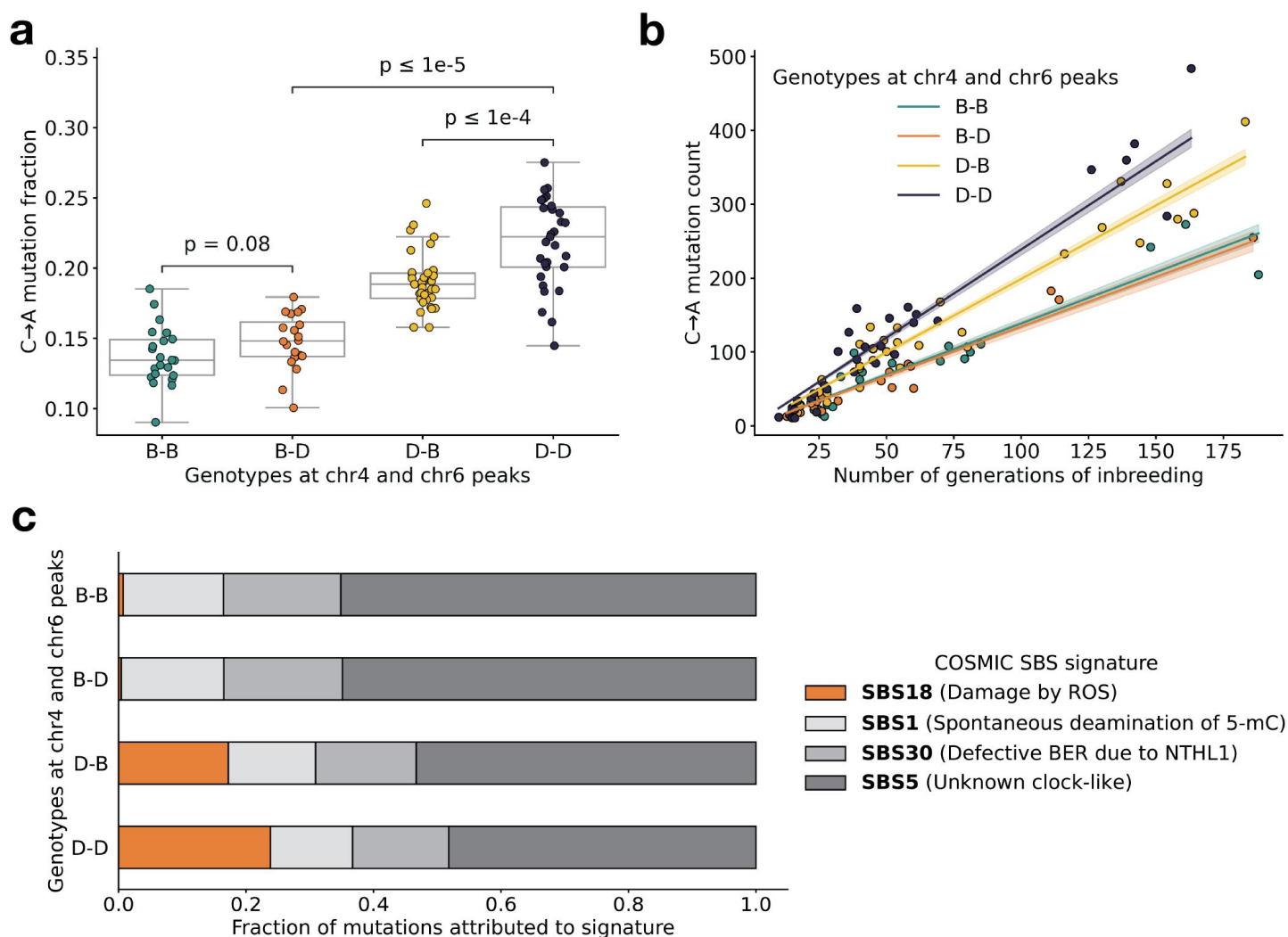


Figure 3

BXD mutation spectra are affected by alleles at both mutator loci.

a) C>A *de novo* germline mutation fractions in BXDs with either *D* or *B* genotypes at markers rs27509845 (chr4 peak) and rs46276051 (chr6 peak). Distributions of C>A mutation fractions were compared with two-sided Mann-Whitney U-tests; annotated p-values are uncorrected. B-B vs. B-D comparison: U-statistic = 149.0, $p = 7.58e-2$; B-D vs D-D comparison: U-statistic = 21.0, $p = 2.61e-8$; D-B vs D-D comparison: U-statistic = 232.5, $p = 6.99e-5$.

b) The count of C>A *de novo* germline mutations in each BXD was plotted against the number of generations for which it was inbred. Lines represent predicted C>A counts in each haplotype group from a generalized linear model (Poisson family, identity link), and shading around each line represents the 95% confidence interval. **c**) Germline mutations in each BXD were assigned to COSMIC SBS mutation signatures using SigProfilerExtractor [27]. After grouping BXDs by their genotypes at rs27509845 and rs46276051 we calculated the fraction of mutations in each group that was attributed to each signature. The proposed etiologies of each mutation signature are: SBS1 (spontaneous deamination of methylated cytosine nucleotides at CpG contexts), SBS5 (unknown, clock-like signature), SBS18 (damage by reactive oxygen species, related to SBS36 and defective base-excision repair due to loss-of-function mutations in MUTYH), and SBS30 (defective base-excision repair due to NTHL1 mutations).

not observe any new mutator loci after performing an IHD scan using BXDs with *B* genotypes at rs27509845 (**Figure 2c**). We also performed QTL scans for the fractions of each 1-mer mutation type using the same mutation data, but none produced a genome-wide significant log-odds score at any locus (**Figure 2-figure supplement 1; Materials and Methods**).

We queried the region surrounding the top marker on chromosome 6 (+/- 5 Mbp) and discovered 63 protein-coding genes, of which five were annotated with a Gene Ontology [23, 24] term related to “DNA repair”: *Fancd2*, *Mbd4*, *Ogg1*, *Rad18*, and *Setmar*. Of these, only three harbored nonsynonymous differences between the parental C57BL/6J and DBA/2J strains (**Table 1**). *Ogg1* encodes a key member of the base-excision repair response to oxidative DNA damage (a pathway that also includes *Mutyh*), *Mbd4* encodes a protein that is involved in the repair of G:T mismatches caused by spontaneous deamination of methylated CpGs, and in mice *Setmar* encodes a SET domain-containing histone methyltransferase.

We also considered the possibility that expression quantitative trait loci (eQTLs), rather than nonsynonymous mutations, could contribute to the C>A mutator phenotype associated with the locus on chromosome 6. Using GeneNetwork [25] we mapped cis-eQTLs for the five aforementioned DNA repair genes in a number of tissues, though we did not have access to expression data from germline cells. *D* alleles near the cosine distance peak on chromosome 6 were significantly associated with decreased *Ogg1* expression in kidney, liver, and gastrointestinal tissues; *D* alleles were also associated with lower *Rad18* expression in the retina (**Table supplement 1**). The cis-eQTLs that lead to decreased *Ogg1* or *Rad18* expression are challenging to interpret, however, given their tissue specificity and our lack of access to germline expression data.

Finally, we queried a dataset of structural variants (SVs) identified via high-quality, long-read assembly of inbred laboratory mouse strains [26] and found 136 large insertions or deletions within 5 Mbp of the cosine distance peak on chromosome 6. Of these, four overlapped the exonic sequences of protein-coding genes (**Table supplement 2**), though none of the genes has a previously annotated role in DNA binding, repair or replication, or in a pathway that would likely affect germline mutation rates. Two protein-coding genes that are involved in DNA repair (*Mbd4* and *Rad18*) harbored intronic insertions or deletions (**Table supplement 2**); however, additional experimental evidence will be needed to probe the functional impact of these SVs.

Evidence of epistasis between germline mutator alleles

Next, we more precisely characterized the effects of the chromosome 4 and 6 mutator alleles on mutation spectra in the BXDs. On average, C>A germline mutation fractions were significantly higher in BXDs with *D* alleles at both mutator loci than in BXDs with *D* alleles at either locus alone (**Figure 3a** and **3-figure supplement 1**). Among BXDs with *B* alleles at the locus overlapping *Mutyh*, those with *D* alleles on chromosome 6 did not exhibit significantly elevated C>A mutation fractions (**Figure 3a**). After controlling for inbreeding duration, we observed that C>A *de novo* mutation counts were always highest in BXDs with *D* alleles at both mutator loci (**Figure 3b**). After 100 generations of inbreeding, BXDs with *D* alleles at both mutator loci were predicted to have 238.9 C>A mutations (95% CI: 231.4 - 246.4), about 20% more than the 199.0 mutations (95% CI: 193.3 - 204.7) predicted in those with *D* and *B* alleles at the chromosome 4 and chromosome 6 loci, respectively (**Figure 3b**).

We also used SigProfilerExtractor [27] to assign the germline mutations in each BXD to single-base substitution (SBS) mutation signatures from the COSMIC catalog [28]. Mutation signatures often reflect specific exogenous or endogenous sources of DNA damage, and the proportions of mutations attributable to particular SBS signatures can suggest a genetic or environmental etiology. The SBS1, SBS5, and SBS30 mutation signatures were active in nearly all BXDs, regardless of genotypes at the chromosome 4 and 6 mutator loci (**Figure 3c**). However, the SBS18 signature, which is dominated by C>A mutations and likely reflects unrepaired DNA damage from

Gene name	Ensembl transcript name	Nucleotide change	Amino acid change	Position in GRCm38/mm10 coordinates	PhyloP conservation score	SIFT prediction
<i>Setmar</i>	ENSMUST00000049246	C>T	p.Leu103Phe	chr6:108,075,853	0.422	0.0 (intolerant/deleterious)

Table 1
Nonsynonymous mutations in DNA repair genes near the chr6 peak

Gene name	Ensembl transcript name	Nucleotide change	Amino acid change	Position in GRCm38/mm10 coordinates	PhyloP conservation score	SIFT prediction
<i>Setmar</i>	ENSMUST00000049246	T>G	p.Ser273Arg	chr6:108,076,365	-0.355	0.3 (tolerant/benign)
<i>Ogg1</i>	ENSMUST00000032406	A>G	p.Thr95Ala	chr6:113,328,510	-0.016	0.84 (tolerant/benign)
<i>Mbd4</i>	ENSMUST00000032469	C>T	p.Asp129Asn	chr6:115,849,644	2.28	0.02 (intolerant/deleterious)

Table 1 (continued)

reactive oxygen species, was almost exclusively active in mice with *D* alleles at the chromosome 4 locus; the highest SBS18 activity was observed in mice with *D* alleles at both mutator loci (**Figure 3c**). SBS18 activity was lowest in mice with *D* alleles at the chromosome 6 mutator locus alone (**Figure 3c**), further demonstrating that *D* alleles at this locus are not sufficient to cause a mutator phenotype.

To more formally test for statistical epistasis, we fit a generalized (Poisson) linear model predicting counts of C>A mutations in each BXD as a function of genotypes at rs27509845 and rs46276051 (the markers with the largest adjusted cosine distance at the two mutator loci); the model also accounted for differences in inbreeding duration and sequencing coverage between the BXDs (*Materials and Methods*). A model that included an interaction term between genotypes at the two markers fit the data significantly better than a model including only additive effects ($p = 7.92e-7$; *Materials and Methods*).

To explore the effects of the two mutator loci in other inbred laboratory mice, we also compared the germline mutation spectra of Sanger Mouse Genomes Project (MGP) strains [29]. Dumont [18] previously identified germline mutations that were private to each of the 29 MGP strains; these private variants likely represent recent *de novo* mutations (**Figure 3**—figure supplement 2). Only two of the MGP strains possess *D* genotypes at both the chromosome 4 and chromosome 6 mutator loci: DBA/1J and DBA/2J. As before, we tested for epistasis in the MGP strains by fitting two linear models predicting C>A mutation counts as a function of genotypes at the two mutator loci. A model incorporating an interaction term did not fit the MGP data significantly better than a model with additive effects alone ($p = 0.806$), so we are unable to confirm the signal of epistasis; however, this may be due to the smaller number of MGP strains with *de novo* germline mutation data.

Some of the candidate mutator alleles are segregating in wild mice

To determine whether the candidate mutator alleles on chromosome 6 were segregating in natural populations, we queried previously published sequencing data generated from 67 wild-derived mice [30]. These data include three subspecies of *Mus musculus*, as well as the outgroup *Mus spretus*. We found that the *Ogg1* *D* allele was segregating at an allele frequency of 0.259 in *Mus musculus domesticus*, the species from which C57BL/6J and DBA/2J derive the majority of their genomes [31], and was fixed in *Mus musculus musculus*, *Mus musculus castaneus*, and the outgroup *Mus spretus* (**Figure 3**—figure supplement 3). The *Setmar* p.Ser273Arg *D* allele was also present at an allele frequency of 0.37 in *Mus musculus domesticus*, while *D* alleles at the *Setmar* p.Leu103Phe variant were not observed in any wild *Mus musculus domesticus* animals. Notably, *D* alleles at the *Mbd4* p.Asp129Asn variant were absent from all wild mouse populations (**Figure 3**—figure supplement 3).

Discussion

Epistasis between germline mutator alleles

We have identified a locus on chromosome 6 that amplifies a C>A germline mutator phenotype in the BXDs, a family of inbred mice derived from the laboratory strains DBA/2J and C57BL/6J. DBA/2J (*D*) alleles at this locus have no detectable effect on C>A mutation rates in mice that also harbor “wild-type” C57BL/6J (*B*) alleles at a previously discovered mutator locus on chromosome 4 [2]. However, mice with *D* alleles at *both* loci have even higher mutation rates than those with *D* alleles at the chromosome 4 mutator locus alone (**Figure 3**). Epistatic interactions between mutator alleles have been previously documented in yeast [32] and in human cell lines [33], but never to our knowledge in a whole-animal context.

Importantly, we discovered epistasis between germline mutator alleles in an unnatural population of model organisms that have been inbred by brother-sister mating in a highly controlled laboratory environment [3]. This breeding setup has likely attenuated the effects of natural selection on all but the most deleterious alleles [19], and may have facilitated the fixation of large-effect mutator alleles that would be less common in wild, outbreeding populations.

We have not conclusively fine-mapped the chromosome 6 mutator locus to a causal variant, but we argue that nonsynonymous mutations in the DNA glycosylases *Ogg1* and *Mbd4* are the best candidates. Our results demonstrate that multiple mutator alleles have spontaneously arisen during the evolutionary history of inbred laboratory mice, supporting the hypothesis that purifying selection is required to keep mutation rates low. Our finding also suggests that mutational pressure can cause mutation rates to rise in just a few generations of relaxed selection in captivity. This hypothesis is corroborated by the recent discovery of a large-effect mutator allele in a rhesus macaque research colony [14], as well as the observation that domesticated animals tend to have higher mutation rates than those in the wild [34].

Possible causal alleles underlying the chromosome 6 mutator locus

Three protein-coding genes involved in DNA repair overlap the C>A mutator locus on chromosome 6 and also contain nonsynonymous differences between the C57BL/6J and DBA/2J founder strains: *Ogg1*, a glycosylase that excises the oxidative DNA lesion 8-oxoguanine (8-oxoG) [4], *Mbd4*, a glycosylase that removes thymine nucleotides at G:T mispairs following spontaneous deamination of methylated CpGs, and *Setmar*, a histone methyltransferase involved in non-homologous end joining (NHEJ) of double-stranded breaks (DSBs) [35, 36].

Both *Ogg1* and *Mbd4* have been previously linked to somatic mutator phenotypes and carcinogenesis. Missense mutations and loss-of-heterozygosity in *Ogg1* have been associated with increased risk of human cancer [37, 38], and copy-number losses of either *Ogg1* or *MutYh* are linked to elevated rates of spontaneous C>A mutation in human neuroblastoma [39]. Biallelic loss-of-function (LOF) mutations in human *MBD4* underlie a neoplastic syndrome that mimics forms of familial adenomatous polyposis caused by LOF mutations in *MUTYH* [40]. Similarly, biallelic loss of *MBD4* function was recently found to cause a hypermutator phenotype in the maternal germline of rhesus macaques [14]; the macaque germline mutator phenotype primarily comprised C>T mutations at CpG and CpA sites, a mutational signature that has previously been associated with LOF mutations in *Mbd4* [41].

Although we are unable to conclusively determine that either *Setmar*, *Ogg1*, or *Mbd4* harbors the causal variant underlying the observed C>A mutator phenotype, we believe that *Ogg1* and *Mbd4* are the most plausible candidates. The primate *SETMAR* ortholog is involved in NHEJ of double-strand breaks, but its role in DNA repair appears to depend on the function of both a SET methyltransferase domain and a *Mariner*-family transposase domain [36, 42, 43]. Since the murine *Setmar* ortholog lacks the latter element, we believe it is unlikely to underlie the epistatic interaction between the chromosome 4 and 6 mutator loci in the BXDs (*Supplementary Information*).

An *Ogg1* mutator allele might impair the excision of 8-oxoguanine lesions

Ogg1 is a member of the same base-excision repair (BER) pathway as *MutYh*, the protein-coding gene we previously implicated as harboring mutator alleles at the locus on chromosome 4 [4]. Each of these genes has a distinct role in the BER response to oxidative DNA damage, and thereby the prevention of C>A mutations [44, 45]. Following damage by reactive oxygen species, *Ogg1* is able to recognize and remove 8-oxoguanine lesions that are base-paired with cytosine

nucleotides; once 8-oxoG is excised, other members of the BER pathway are mobilized to restore a proper G:C base pair at the site. If an 8-oxoG lesion is not removed before the cell enters S-phase, adenine can be mis-incorporated opposite 8-oxoG during DNA replication [44]. If this occurs, *Mutyh* can excise the mispaired adenine, leaving a one-nucleotide gap that is processed and filled with a cytosine by other BER proteins. The resulting C:8-oxoG base pair can then be “returned” to *Ogg1* for excision and repair. Defects in the BER response to oxidative damage lead to significantly elevated rates of C>A mutation. For example, triple-knockout (KO) mice lacking *Ogg1*, *Mutyh*, and *Mth1* (which encodes an enzyme that prevents 8-oxo-dGTP from being incorporated during DNA synthesis [46]) accumulate a 100-fold excess of 8-oxoG in their gonadal cells [45]. Almost 99% of *de novo* germline mutations in the *Ogg1/Mutyh/Mth1* triple KO mice are C>A transversions, demonstrating the clear role of 8-oxoG repair in preventing C>A mutation.

The p.Thr95Ala *Ogg1* missense variant is not predicted to be deleterious by the *in silico* tool SIFT [47], and occurs at a nucleotide that is not particularly well-conserved across mammalian species (Table 1). We also observe that the *D* allele at p.Thr95Ala is segregating at an allele frequency of approximately 26% among wild-derived *Mus musculus domesticus* animals, and is fixed in other wild populations of *Mus musculus musculus*, *Mus musculus castaneus*, and *Mus spretus*. Although we would expect *a priori* that *Ogg1* deficiency should lead to increased 8-oxoG accumulation and elevated C>A mutation rates, these lines of evidence suggest that p.Thr95Ala is not highly deleterious on its own, and might only exert a detectable effect on the BER gene network when *Mutyh* function is also impaired. It is also possible that *D* alleles at *Ogg1* lead to a very subtle increase in C>A mutation rates, and we are simply underpowered to detect such a small mutation rate effect in the BXDs. Overall, it is challenging to predict the functional consequences of the *Ogg1* p.Thr95Ala variant, and as we discuss below, the p.Asp129Asn missense mutation in *Mbd4* may be an even more compelling candidate mutator allele.

***Mbd4* may buffer the effects of *Mutyh* mutator alleles by triggering apoptosis**

Unlike the *Ogg1* p.Thr95Ala mutation, the p.Asp129Asn variant in *Mbd4* resides within an annotated protein domain (the *Mbd4* methyl-CpG binding domain), occurs at a nucleotide and amino acid residue that are both well-conserved, and is predicted to be deleterious by SIFT [47] (Table 1). A missense mutation that affects the homologous amino acid in humans (p.Asp142Gly in GRCh38/hg38) is also present on a single haplotype in the Genome Aggregation Database (gnomAD) [48] and is predicted by SIFT and Polyphen [49] to be “deleterious” and “probably_damaging” in human genomes, respectively.

One puzzling observation is that loss-of-function mutations in *Mbd4* are not typically associated with C>A mutator phenotypes. Instead, *Mbd4* deficiency is usually implicated in C>T mutagenesis at CpG sites, and we did not detect an excess of C>T mutations in BXDs with *D* alleles at the chromosome 6 mutator locus (Figure 3-figure supplement 1). However, loss of function mutations in *Mbd4* have also been shown to exacerbate the effects of exogenous DNA damage agents. For example, mouse embryonic fibroblasts that harbor homozygous *Mbd4* knockouts fail to undergo apoptosis following treatment with a number of chemotherapeutics and mutagenic compounds [50]. Most of these exogenous mutagens cause DNA damage that is normally repaired by mismatch repair (MMR) machinery, but murine intestinal cells with biallelic *Mbd4* LOF mutations also showed a reduced apoptotic response to gamma irradiation, which is repaired independently of the MMR gene *Mlh1* [51]. Homozygous loss of *Mbd4* function also leads to accelerated intestinal tumor formation in mice that harbor an *Apc* allele that predisposes them to intestinal neoplasia [41], and mice with biallelic truncations of the *Mbd4* coding sequence exhibit modestly increased mutation rates in colon cancer cell lines, including increased C>A mutation rates in certain lines [52].

Together, these lines of evidence suggest that *Mbd4* can modulate sensitivity to many types of exogenous mutagens, potentially through its role in determining whether cells harboring DNA damage should undergo apoptosis [50, 51]. We speculate that in mice with deficient 8-oxoguanine repair — caused by a mutator allele in *Mutyh*, for example — reactive oxygen species (ROS) could cause accumulation of DNA damage in the germline. If those germ cells harbor fully functional copies of *Mbd4*, they might be able to trigger apoptosis and partially mitigate the effects of a *Mutyh* mutator allele. However, mice with reduced activity of both *Mbd4* and *Mutyh* may have a reduced ability to initiate cell death in response to DNA damage; as a result, their germ cells may accumulate even higher levels of ROS-mediated damage, leading to substantially elevated germline C>A mutation rates.

Our model makes the testable prediction that mice with *D* haplotypes at *Mbd4* should have elevated sensitivity to a variety of mutagenic agents. We thus anticipate that future experimental work will be able to more conclusively establish a mechanistic explanation for the epistatic interaction between mutator loci described in this paper.

No indication of causal structural variation or mobile element insertions near the chromosome 6 mutator locus

Although we argue above that *Mbd4* and *Ogg1* are the best candidate genes to explain the augmented C>A mutator phenotype in a subset of BXDs, we cannot conclusively determine that either the p.Asp129Asn or p.Thr95Ala missense mutation is a causal allele. We previously hypothesized that *Mutyh* missense mutations on *D* haplotypes were responsible for the large-effect C>A mutator phenotype we observed in the BXDs [2]. However, subsequent long-read assemblies of several inbred laboratory mouse strains revealed that this mutator phenotype might be caused by a ~5 kbp mobile element insertion (MEI) within the first intron of *Mutyh* [26], which is associated with significantly reduced expression of *Mutyh* in embryonic stem cells. We queried the new high-quality assemblies for evidence of mobile elements or other large structural variants (SVs) in the region surrounding the mutator locus on chromosome 6, but found no similarly compelling evidence that either SVs or MEIs might underlie the mutator phenotype described in this study.

Strengths and limitations of the inter-haplotype distance approach

Our inter-haplotype distance (IHD) approach was able to identify a mutator allele that escaped notice using quantitative trait locus (QTL) mapping. To more systematically compare the power of IHD and QTL mapping, we performed simulations under a variety of possible parameter regimes. Overall, we found that IHD and QTL mapping have similar power to detect mutator alleles on haplotypes that each harbor tens or hundreds of *de novo* germline mutations (Figure 1-figure supplement 2). Nonetheless, only IHD was able to discover the mutator locus on chromosome 6 in the BXDs, demonstrating that it outperforms QTL mapping in certain experimental systems (for example, in which RILs have been inbred for different lengths of time and carry varying numbers of mutations that can be leveraged for mutator mapping). Because the BXDs were generated in six breeding epochs over a period of nearly 40 years, the oldest lines have accumulated orders of magnitude more mutations than the youngest lines; these younger BXDs have much noisier mutation spectra as a result. While approaches for QTL mapping typically weight the phenotypic measurements of each sample equally, IHD compares the *aggregate* mutation spectra of haplotypes at every locus, a property that likely increased its power to detect mutators in the BXD dataset.

Another benefit of the IHD approach is that it obviates the need to perform separate association tests for every possible *k*-mer mutation type, and therefore the need to adjust significance thresholds for multiple tests. Since IHD compares the complete mutation spectrum between

haplotypes that carry either allele at a site, it would also be well-powered to detect a mutator allele that exerted a coordinated effect on multiple *k*-mer mutation types (e.g., increased the rates of both C>T and C>A mutations).

However, the IHD method suffers a handful of drawbacks when compared to QTL mapping. Popular QTL mapping methods (such as R/qt12 [53]) use linear models to test associations between genotypes and phenotypes, enabling the inclusion of additive and interactive covariates, as well as kinship matrices, in QTL scans. Although we have developed methods to account for inter-sample relatedness in the IHD approach (*Materials and Methods*), they are not as flexible as similar methods in QTL mapping software. Additionally, the IHD method assumes that mutator alleles affect a subset of *k*-mer mutation types; if a mutator allele increased the rates of all mutation types equally on haplotypes that carried it, IHD would be unable to detect it.

Discovering mutator alleles in other experimental systems

Our discovery of a second BXD mutator allele underscores the power of recombinant inbred lines (RILs) as a resource for dissecting the genetic architecture of germline mutation rates. Large populations of RILs exist for many model organisms, and we anticipate that as whole-genome sequencing becomes cheaper and cheaper, the IHD method could be useful for future mutator allele discovery outside of the BXDs. At the same time, RILs are a finite resource that require enormous investments of time and labor to construct. If germline mutator alleles are only detectable in these highly unusual experimental populations, we are unlikely to discover more than a small fraction of the mutator alleles that may exist in nature.

Fortunately, the approach introduced in this paper is readily adaptable to datasets beyond RILs. Thousands of human pedigrees have been sequenced in an effort to precisely estimate the rate of human *de novo* germline mutation [6, 7, 54], and as family sequencing has become a more common step in the diagnosis of many congenital disorders, these datasets are growing on a daily basis. Large cohorts of two- or three-generation families are an example of a regime in which IHD could enjoy high power; by pooling sparse mutation counts across many individuals who share the same candidate mutator allele, even a subtle mutator signal could potentially rise above the noise of *de novo* germline mutation rate estimates.

Selection on germline mutator alleles will likely prevent large-effect mutators from reaching high allele frequencies, but a subset may be detectable by sequencing a sufficient number of human trios [55]. Since germline mutators often seem to exert their effects on a small number of mutation types, mutation spectrum analyses may have greater power to detect the genes that underlie heritable mutation rate variation, even if each gene has only a modest effect on the overall mutation rate per generation.

Materials and Methods

Identifying *de novo* germline mutations in the BXDs

The BXD resource currently comprises a total of 152 recombinant inbred lines (RILs). BXDs were derived from either F2 or advanced intercrosses, and subsequently inbred by brother-sister mating for up to 180 generations [3]. BXDs were generated in distinct breeding “epochs,” which were each initiated with a distinct cross of C57BL/6J and DBA/2J parents; epochs 1, 2, 4, and 6 were derived from F2 crosses, while epochs 3 and 5 were derived from advanced intercrosses [3]. Previously, we analyzed whole-genome sequencing data from the BXDs and identified candidate *de novo* germline mutations in each line [2]. A detailed description of the methods used for DNA extraction, sequencing, alignment, and variant processing, as well as the characteristics of the *de novo* mutations, are available in a previous manuscript [2].

Briefly, we identified private single-nucleotide mutations in each BXD that were absent from all other BXDs, as well as from the C57BL/6J and DBA/2J parents. We required each private variant to be meet the following criteria:

- genotyped as either homozygous or heterozygous for the alternate allele, with at least 90% of sequencing reads supporting the alternate allele
- supported by at least 10 sequencing reads
- Phred-scaled genotype quality of at least 20
- must not overlap regions of the genome annotated as segmental duplications or simple repeats in GRCm38/mm10
- must occur on a parental haplotype that was inherited by at least one other BXD at the same locus; these other BXDs must be homozygous for the reference allele at the variant site

A new approach to discover germline mutator alleles

Calculating inter-haplotype distance

We developed a new approach to discover loci that affect the germline *de novo* mutation spectrum in biparental RILs ([Figure 1](#)).

We assume that a collection of haplotypes has been genotyped at informative markers, and that *de novo* germline mutations have been identified on each haplotype.

At each informative marker, we divide haplotypes into two groups based on the parental allele that they inherited. We then compute a *k*-mer mutation spectrum using the aggregate mutation counts in each haplotype group. The *k*-mer mutation spectrum contains the frequency of every possible *k*-mer mutation type in a collection of mutations, and can be represented as a vector of size $6 \times 4^{k-1}$ after collapsing by strand complement. For example, the 1-mer mutation spectrum is a 6-element vector that contains the frequencies of C>T, C>G, C>A, A>G, A>T, and A>C mutations. Since C>T transitions at CpG nucleotides are often caused by a distinct mechanism (spontaneous deamination of methylated cytosine), we expand the 1-mer mutation spectrum to include a separate category for CpG>TpG mutations [[56](#)].

At each marker, we then compute the cosine distance between the two aggregate spectra. The cosine distance between two vectors **A** and **B** is defined as

$$D^C = 1 - \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

Where $\|\mathbf{A}\|$ and $\|\mathbf{B}\|$ are the L^2 (or Euclidean) norms of **A** and **B**, respectively. The cosine distance metric has a number of favorable properties for comparing mutation spectra. Since it adjusts for the magnitude of the two input vectors, cosine distance can be used to compare two spectra with unequal total mutation counts (even if those total counts are relatively small). Additionally, by calculating the cosine distance between mutation spectra, we avoid the need to perform separate comparisons of mutation counts at each individual *k*-mer mutation type.

Inspired by methods from QTL mapping [53, 57], we use permutation tests to establish genome-wide cosine distance thresholds. In each of N permutation trials, we randomly shuffle the per-haplotype mutation data such that haplotype labels no longer correspond to the correct mutation counts. Using the shuffled mutation data, we perform a genome-wide scan as described above, and record the maximum cosine distance observed at any locus. After N permutations (usually 10,000), we compute the 1 - p percentile of the distribution of maximum statistics, and use that percentile value as a genome-wide significance threshold (for example, at $p = 0.05$).

Accounting for relatedness between strains

We expect each BXD to derive approximately 50% of its genome from C57BL/6J and 50% from DBA/2J. As a result, every pair of BXDs will likely have identical genotypes at a fraction of markers. Pairs of more genetically similar BXDs may also have more similar mutation spectra, potentially due to shared polygenic effects on the mutation process. Therefore, at a given marker, if the BXDs that inherited D alleles are more genetically dissimilar from those that inherited B alleles (considering all loci throughout the genome in our measurement of genetic similarity), we might expect the aggregate mutation spectra in the two groups to also be more dissimilar.

We implemented a simple approach to account for these potential issues of relatedness. At each marker g_i , we divide BXD haplotypes into two groups based on the parental allele they inherited. As before, we first compute the aggregate mutation spectrum in each group of haplotypes and calculate the cosine distance between the two aggregate spectra (D_i^C). Then, within each group of haplotypes, we calculate the allele frequency of the D allele at every marker along the genome to obtain a vector of length n , where n is the number of genotyped markers. To quantify the genetic similarity between the two groups of haplotypes, we calculate the Pearson correlation coefficient r_i between the two vectors of marker-wide D allele frequencies.

Put another way, at every marker g_i along the genome, we divide BXD haplotypes into two groups and compute two metrics: D_i^C (the cosine distance between the two groups' aggregate spectra) and r_i (the correlation between genome-wide D allele frequencies in the two groups). To control for the potential effects of genetic similarity on cosine distances, we regress $D_1^C, D_2^C, \dots, D_n^C$ on $(r_1, r_2, \dots, r_n)$ for all n markers using an ordinary least-squares model. We then use the residuals from the fitted model as the "adjusted" cosine distance values for each marker. If genome-wide genetic similarity between haplotypes perfectly predicts cosine distances at each marker, these residuals will all be 0 (or very close to 0). If genome-wide genetic similarity has no predictive power, the residuals will simply represent the difference between the observed cosine distance at a single marker and the marker-wide mean of cosine distances.

Accounting for BXD population structure due to breeding epochs

The current BXD family was generated in six breeding "epochs." As discussed previously, each epoch was initiated with a distinct cross of C57BL/6J and DBA/2J parents; BXDs in four of the epochs were generated following F2 crosses of C57BL/6J and DBA/2J, and BXDs in the other two were generated following advanced intercrosses. Due to this breeding approach the BXD epochs differ from each other in a few important ways. For example, BXDs derived in epochs 3 and 5 (i.e., from advanced intercross) harbor larger numbers of fixed recombination breakpoints than those from epochs 1, 2, 4, and 6 [3]. Although the C57BL/6J and DBA/2J parents used to initialize each epoch were completely inbred, they each possessed a small number unique *de novo* germline mutations that were subsequently inherited by many of their offspring. A number of these "epoch-specific" variants have also been linked to phenotypic variation observed between BXDs from different epochs [3, 58, 59, 60].

To account for potential population structure, as well as these epoch-specific effects, we introduced the ability to perform stratified permutation tests in the inter-haplotype distance approach. Normally, in each of N permutations we shuffle the per-haplotype mutation spectrum

data such that haplotype labels no longer correspond to the correct mutation spectra (i.e., shuffle mutation spectra *across* epochs). In the stratified approach, we instead shuffle per-haplotype mutation data *within* epochs, preserving epoch structure while still enabling mutation spectra permutations.

Implementation and source code

The inter-haplotype distance method was implemented in Python, and relies heavily on the following Python libraries: numpy, pandas, matplotlib, scikit-learn, pandera, seaborn, and numba [61 [62](#), 63 [64](#), 65 [66](#), 67 [68](#)].

The code underlying IHD, as well as documentation of the method, is available on GitHub (<https://github.com/quinlan-lab/proj-mutator-mapping>). We have also deposited a reproducible Snakemake [68 [69](#)] workflow for running reproducing all analyses and figures presented in the manuscript.

Simulations to assess the power of the inter-haplotype distance approach

We performed a series of simple simulations to estimate our power to detect alleles that affect the germline mutation spectrum using the inter-haplotype distance method.

Simulating genotypes

First, we simulate genotypes on a population of haplotypes at a collection of sites. We define a matrix of size (s, h) , where s is the number of sites and h is the number of haplotypes. We assume that every site is biallelic, and that the minor allele frequency at every site is 0.5. For every entry $G_{i,j}$, we take a single draw from a uniform distribution in the interval $[0,0.1,0]$. If the value of that draw is less than 0.5, we assign the value of $G_{i,j}$ to be 1. Otherwise, we assign the value of $G_{i,j}$ to be 0.

Defining expected mutation type probabilities

Next, we define a vector of 1-mer mutation probabilities:

$$P = (0.29, 0.17, 0.12, 0.075, 0.1, 0.075, 0.17)$$

These probabilities sum to 1 and roughly correspond to the expected frequencies of C>T, CpG>TpG, C>A, C>G, A>T, A>C, and A>G *de novo* germline mutations in mice, respectively [10 [11](#)]. If we are simulating the 3-mer mutation spectrum, we modify the vector of mutation probabilities P to be length 96, and assign every 3-mer mutation type a value of $\frac{P_c}{16}$, where P_c is the probability of the “central” mutation type associated with the 3-mer mutation type. In other words, each of the 16 possible N_CN>N_TN 3-mer mutation types would be assigned a mutation probability of $\frac{P_c}{16} = \frac{0.46}{16} = 0.02875$. We then generate a vector of lambda values by scaling the mutation probabilities by the number of mutations we wish to simulate (m):

$$\lambda = Pm$$

We also create a second vector of lambda values ($\times$), in which we multiply the $\times$ value of a single mutation type by the mutator effect size e .

In our simulations, we assume that genotypes at a single site (the “mutator locus”) are associated with variation in the mutation spectrum. That is, at a single site S_i , all of the haplotypes with 1 alleles should have elevated rates of a particular mutation type and draw their mutation counts from λ , while all of the haplotypes with 0 alleles should have “wild-type” rates of that mutation type and draw their mutation counts from λ . We therefore pick a random site to be the “mutator locus,” and identify the indices of haplotypes in G that were assigned 1 alleles at S_i . We call these indices h_{mut} .

Simulating mutation spectra

To simulate the mutation spectrum on our toy population of haplotypes, we define a matrix C of size (h, n) , where $n = 6 \times 4^{k-1}$ (or if $k = 1$ and we include CpG>TpG mutations, $6 \times 4^{k-1} + 1$).

Then, we populate the matrix C separately for *mutator* and *wild-type* haplotypes. For every row i in the matrix (i.e., for every haplotype), we first ask if i is in h_{mut} (that is, if the haplotype at index i was assigned a 1 allele at the “mutator locus”). If so, we set the values of C_i to be the results of a single Poisson draw from λ . If row i is not in h_{mut} , we set the values of C_i to be the results of a single Poisson draw from λ .

Assessing power to detect a simulated mutator allele using IHD

For each combination of parameters (number of simulated haplotypes, number of simulated markers, mutator effect size, etc.), we run 100 independent trials. In each trial, we simulate the genotype matrix G and the mutation counts C . We calculate a “focal” cosine distance as the cosine distance between the aggregate mutation spectra of haplotypes with either genotype at s_i (the site at which we artificially simulated an association between genotypes and mutation spectrum variation). We then perform an inter-haplotype distance scan using $N = 1,000$ permutations. If fewer than 5% of the N permutations produced a cosine distance greater than or equal to the focal distance, we say that the approach successfully identified the mutator allele in that trial.

Assessing power to detect a simulated mutator allele using quantitative trait locus (QTL) mapping

Using simulated data, we also assessed the power of traditional quantitative trait locus (QTL) mapping to detect a locus associated with mutation spectrum variation. As described above, we simulated both genotype and mutation spectra for a population of haplotypes under various conditions (number of mutations per haplotype, mutator effect size, etc.). Using those simulated data, we used R/qtl2 [53] to perform a genome scan for significant QTL as follows; we assume that the simulated genotype markers are evenly spaced (in physical Mbp coordinates) on a single chromosome. First, we calculate the fraction of each haplotype’s *de novo* mutations that belong to each of the $6 \times 4^{k-1}$ possible k -mer mutation types. We then convert the simulated genotypes at each marker to genotype probabilities using the `calc_genoprob` function in R/qtl2, with `map_function` × “c-f” and `error_prob` = 0. For every k -mer mutation type, we use genotype probabilities and per-haplotype mutation fractions to perform a scan for QTL with the `scan1` function; to make the results more comparable to those from the IHD method, we do not include any covariates or kinship matrices in these QTL scans. We then use the `scan1perm` function to perform 1,000 permutations of the per-haplotype mutation fractions and calculate log-odds (LOD) thresholds for significance. We consider the QTL scan to be “successful” if it produces a LOD score above the significance threshold (defined using $\alpha = \frac{0.05}{7}$) for the marker at which we simulated an association with mutation spectrum variation.

Applying the inter-haplotype distance method to the BXDs

We downloaded previously-generated BXD *de novo* germline mutation data from the GitHub repository associated with our previous manuscript, which was also archived at Zenodo [25, 69, 70], and downloaded a CSV file of BXD genotypes at ~7,300 informative markers from GeneNetwork [25, 71]. We also downloaded relevant metadata about each BXD from the manuscript describing the updated BXD resource [3]. These files are included in the GitHub repository associated with this manuscript.

As in our previous manuscript [2], we included mutation data from a subset of the 152 BXDs in our inter-haplotype distance scans. Specifically, we removed BXDs that were backcrossed to a C57BL/6J or DBA/2J parent at any point during the inbreeding process (usually, in order to rescue that BXD from inbreeding depression [3]). We also removed BXD68 from our genome-wide scans, since we previously discovered a hyper-mutator phenotype in that line; the C>A germline mutation rate in BXD68 is over 5 times the population mean, likely due to a private deleterious nonsynonymous mutation in *Mutyh* [2]. In our previous manuscript, we removed any BXDs that had been inbred for fewer than 20 generations, as it takes approximately 20 generations of strict brother-sister mating for an RIL genome to become >98% homozygous [72]. As a result, any potential mutator allele would almost certainly be either fixed or lost after 20 generations; if fixed, the allele would remain linked to any excess mutations it causes for the duration of subsequent inbreeding. In other words, the *de novo* mutations present in the genome of a “young” BXD (i.e., a BXD that was inbred for fewer than 20 generations) would not reflect a mutator allele’s activity as strongly as the mutations present in the genome of a much older BXD. This presented a challenge when we used quantitative trait locus mapping to discover mutator alleles in our previous manuscript, since the phenotypes (i.e., C>A mutation rates) of young and old BXDs were weighted equally; thus, we simply removed the younger BXDs from our analysis to avoid using their especially noisy mutation spectra. Since IHD computes an *aggregate* mutation spectrum using all BXDs that inherited a particular allele at a locus, and can overcome the sparsity and noise of individual mutation spectra, we chose to include these younger BXDs in our genome-wide scans in this study.

In total, we included 117 BXDs in our genome-wide scans.

Identifying candidate single-nucleotide mutator alleles overlapping the chromosome 6 peak

We investigated the region implicated by our inter-haplotype distance approach on chromosome 6 by subsetting the joint-genotyped BXD VCF file (European Nucleotide Archive accession PRJEB45429 [73]) using bcftools [74]. We defined the candidate interval surrounding the cosine distance peak on chromosome 6 as +/- 5 Mbp from the genotype marker with the largest adjusted cosine distance value (rs46276051). To predict the functional impacts of both single-nucleotide variants and indels on splicing, protein structure, etc., we annotated variants in the BXD VCF using the following snpEff [75] command:

```
java -Xmx16g -jar /path/to/snpEff/jarfile GRCm38.75 /path/to/bxd/vcf >
/path/to/uncompressed/output/vcf
```

and used cyvcf2 [76] to iterate over the annotated VCF file in order to identify nonsynonymous fixed differences between the parental C57BL/6J and DBA/2J strains.

Identifying candidate structural variant alleles overlapping the chromosome 6 peak

We downloaded summary VCFs containing insertion, deletion and inversion structural variants (identified via high-quality, long-read assembly of inbred laboratory mouse strains [26]) from the Zenodo link associated with the Ferraj et al. manuscript: <https://doi.org/10.5281/zenodo.7644286>.

We then downloaded a TSV file containing RefSeq gene predictions in GRCm39/mm39 from the UCSC Table Browser [77], and used the bx-python library [78] to intersect the interval spanned by each structural variant with the intervals spanned by the txStart and txEnd of every RefSeq entry.

We queried all structural variants within a region +/- 5 Mbp from the adjusted cosine distance peak on chromosome 6 at marker rs46276051.

Extracting mutation signatures

We used SigProfilerExtractor (v.1.1.21) [28] to extract mutation signatures from the BXD mutation data. After converting the BXD mutation data to the “matrix” input format expected by SigProfilerExtractor, we ran the sigProfilerExtractor method as follows:

```
# install the mm10 mouse reference data
genInstall.install('mm10')

# run mutation signature extraction
sig.sigProfilerExtractor(
    'matrix',
    /path/to/output/directory,
    /path/to/input/mutations,
    maximum_signatures=10,
    nmf_replicates=100,
    opportunity_genome="mm10",
)
```

Comparing mutation spectra between Mouse Genomes Project strains

We downloaded mutation data from a previously published analysis [18] (Supplementary File 1, Excel Table S3) that identified strain-private mutations in 29 strains that were originally whole-genome sequenced as part of the Sanger Mouse Genomes (MGP) project [29]. When comparing counts of each mutation type between MGP strains that harbored either *D* or *B* alleles at the chromosome 4 or chromosome 6 mutator loci, we adjusted mutation counts by the number of callable A, T, C, or G nucleotides in each strain as described previously [2].

Querying GeneNetwork for eQTLs at the mutator locus

We used the online GeneNetwork resource [25], which contains array- and RNA-seq-derived expression measurements in a wide variety of tissues, to find *cis*-eQTLs for the DNA repair genes we implicated under the cosine distance peak on chromosome 6. On the GeneNetwork homepage (genenetwork.org), we selected the “BXD Family” **Group** and used the **Type** dropdown menu to select each of the specific expression datasets described in, **Table 2**. In the **Get Any** text box, we

then entered the listed gene name and clicked **Search**. After selecting the appropriate trait ID on the next page, we used the **Mapping Tools** dropdown to run Haley-Knott regression [79] with the following parameters: WGS-based marker genotypes, 1,000 permutations for LOD threshold calculations, and controlling for BXD genotypes at the rs32497085 marker.

The exact names of the expression datasets we used for each tissue are shown in **Table 2** below:

Calculating the frequencies of candidate mutator alleles in wild mice

To determine the frequencies of the *Ogg1*, *Mbd4*, and *Setmar* nonsynonymous mutations in other populations of mice, we queried a VCF file containing genome-wide variation in 67 wild-derived mice from four species of *Mus* [30]. We calculated the allele frequency of each nonsynonymous mutation in each of the four species or subspecies (*Mus musculus domesticus*, *Mus musculus musculus*, *Mus musculus castaneus*, and *Mus spretus*), including genotypes that met the following criteria:

- supported by at least 10 sequencing reads
- Phred-scaled genotype quality of at least 20

Testing for epistasis between the two mutator loci

To test for statistical epistasis between the mutator loci on chromosome 4 and chromosome 6, we modeled C>A mutation rates in the BXDs as a function of genotypes at either locus. Specifically, we tested for statistical interaction between genotypes by fitting a generalized linear model in the R statistical language as follows:

```
m1 <- glm(Count ~ offset(log(ADJ_AGE)) + Genotype_A * Genotype_B, data =
  data, family=poisson())
```

In this model, Count is the count of C>A *de novo* mutations observed in each BXD. ADJ_AGE is the product of the number of “callable” cytosine/guanine nucleotides in each BXD (i.e., the total number of cytosines/guanines covered by at least 10 sequencing reads) and the number of generations for which the BXD was inbred. We included the logarithm of ADJ_AGE as an “offset” in order to model the response variable as a rate (expressed per base-pair, per generation) rather than an absolute count; the BXDs differ in both their durations of inbreeding and the proportions of their genomes that were sequenced to sufficient depth, which influences the number of mutations we observe in each BXD. The Genotype_A and Genotype_B terms represent the genotypes of BXDs at markers rs27509845 and rs46276051 (the markers with peak cosine distances on chromosomes 4 and 6 in the two inter-haplotype distance scans). We limited our analysis to the n = 108 BXDs that were homozygous at both sites, allowing us to model genotypes at either locus as binary variables (“B” or “D”). Using analysis of variance (ANOVA), we then compared the model including an interaction effect to a model including only additive effects:

```
m2 <- glm(Count ~ offset(log(ADJ_AGE)) + Genotype_A + Genotype_B, data =
  data, family=poisson())
```

Tissue name	Complete name of GeneNetwork expression data
Kidney	Mouse kidney M430v2 Sex Balanced (Aug06) RMA
Gastrointestinal	UTHSC Mouse BXD Gastrointestinal Affy MoGene 1.0 ST Gene Level (Apr14) RMA
Hematopoietic stem cells	UMCG Stem Cells ILM6v1.1 (Apr09) transformed

Table 2
Names of gene expression datasets used for each tissue type on GeneNetwork

Tissue name	Complete name of GeneNetwork expression data
Hematopoietic progenitor cells	UMCG Progenitor Cells ILM6v1.1 (Apr09) transformed
Spleen	UTHSC Affy MoGene 1.0 ST Spleen (Dec10) RMA
Liver	UTHSC BXD Liver RNA-Seq Avg (Oct19) TPM Log2
Heart	NHLBI BXD All Ages Heart RNA-Seq (Nov20) TMP Log2 **
Eye	UTHSC BXD All Ages Eye RNA-Seq (Nov20) TPM Log2 **

Table 2 (continued)

```
anova(m1, m2, test="Chisq")
```

We tested for epistasis in the Sanger Mouse Genomes Project (MGP) strains using a nearly-identical approach. In this analysis, we fit two models as follows:

```
m1 <- glm(Count ~ offset(log(CALLABLE_C)) + Genotype_A * Genotype_B, data =  
  data, family=poisson())  
  
m2 <- glm(Count ~ offset(log(CALLABLE_C)) + Genotype_A + Genotype_B, data =  
  data, family=poisson())
```

where Count is the count of strain-private C>A mutations observed in each MGP strain [18]. The CALLABLE_C term represents the total number of cytosine and guanine nucleotides that were accessible for mutation calling in each strain, and the Genotype_A and Genotype_B terms represent MGP genotypes at the chromosome 4 and chromosome 6 mutator loci, respectively. We compared the two models using ANOVA as described above.

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Reviewer #1 (Public Review):

The mutation rate and spectrum have been found to differ between populations as well as across individuals within the same population. Hypothesizing that some of the observed variation has a genetic basis, the authors of this paper have made important contributions in the past few years in identifying genetic variants that modify mutation rate or spectrum in natural populations. This paper makes one significant step further by developing a new method for mapping genetic variants associated with the mutation spectrum, which reveals new biological insights.

Using traditional quantitative trait locus (QTL) mapping in the BXD mouse recombinant inbred lines (RILs), the authors of this paper previously identified a genetic locus associated with C>A mutation rate. However, this approach has limited power, as it suffers from multiple testing burden as well as noise in the "observed mutation rate/spectrum phenotype" due to rarity and randomness of mutation events. To overcome these limitations, the authors developed a new method that they named "inter-haplotype distance" (IHD), which in short measures the difference in the aggregate mutation spectrum between two groups of individuals with distinct genotypes at a specific genomic locus. With this new approach, they recover the previously reported candidate mutator locus (near *Mutyh* gene) and identify a new candidate variant that modifies the C>A mutation rate on only one genetic background. Using more rigorous statistical testing, the authors show convincingly synergistic epistatic effects between the mutator alleles at the two loci.

Overall, the analyses presented are well done and provide convincing evidence for the major findings, including the new candidate mutator locus and its epistatic interaction with the *Mutyh* locus. The new IHD method introduced is innovative and outperforms traditional QTL mapping under certain conditions, but some of its statistical properties and limitations are not fully described. The part that describes how the method works is a little hard to follow (partially due to the confusing name; see comments below), but the rest of the paper is very

well written. Below are my comments and suggestions on how to improve, but I identify no major issues.

The name of the new method "inter-haplotype distance" is more confusing than helpful, as the haplotype information is not critical for implementing this method. First, the mutation spectrum is aggregated genome-wide regardless of the haplotypes where the mutations are found. Second, the only critical haplotype information is that at the focal site (i.e., the locus that is tested for association): individuals are aggregated together when they belong to the same "haplotype group" at the focal site. However, for the classification step, haplotype information is not really necessary: individuals can be grouped based on their genotypes at the given locus (e.g., AA vs AB). As the authors mentioned, this method can be potentially applied to other mutation datasets, where haplotype information may well be unavailable. I hope the authors can reconsider the name and remove the term "haplotype" (perhaps something like "inter-genotype distance"?) to avoid giving the wrong impression that haplotype information is critical for applying this method.

The biggest advantage of the IHD method over QTL mapping is alleviation of the multiple testing burden, as one comparison tests for any changes in the mutation spectrum, including simultaneous, small changes in the relative abundance of multiple mutation types. Based on this, the authors claim that IHD is more powerful to detect a mutator allele that affects multiple mutation types. Although logically plausible, it is unclear under what quantitative conditions IHD can actually have greater power over QTL. It will be helpful to support this claim by providing some simulation results.

The flip side of this advantage of IHD is that, when a significant association is detected, it is not immediately clear which mutation type is driving the signal. Related to this, it is unclear how the authors reached the point that "...the C>A mutator phenotype associated with the locus on chromosome 6", when they only detected significant IHD signal at rs46276051 (on Chr6), when conditioning on D genotypes at the rs27509845 (on Chr4) and no significant signal for any 1-mer mutation type by traditional mapping. The authors need to explain how they deduced that C>A mutation is the major source of the signal. In addition, beyond C>A mutations, can mutation types other than C>A contribute to the IHD signal at rs46276051? More generally, I hope the authors can provide some guidelines on how to narrow a significant IHD signal to specific candidate mutation type(s) affected, which will make the method more useful to other researchers.

To account for differential relatedness between the inbred lines, the authors regressed the cosine distance between the two aggregate mutation spectra on the genome-wide genetic similarity and take the residual as the adjusted test metric. What is the value of the slope from this regression? If significantly non-zero, this would support a polygenic architecture of the mutation spectrum phenotype, which could be interesting. If not, is this adjustment really necessary? In addition, is the intercept assumed to be zero for this regression, and does such an assumption matter? I would appreciate seeing a supplemental figure on this regression.

Reviewer #2 (Public Review):

In this paper Sasani, Quinlan and Harris present a new method for identifying genetic factors affecting germline mutation, which is particularly applicable to genome sequence data from mutation accumulation experiments using recombinant inbred lines. These are experiments where laboratory organisms are crossed and repeatedly inbred for many generations, to build up a substantial number of identifiable germline mutations. The authors apply their method to such data from mice, and identify two genetic factors at two separate genetic loci. Clear evidence of such factors has been difficult to obtain, so this is an important finding. They further show evidence of an epistatic interaction between these factors (meaning that they do not act independently in their effects on the germline mutation process). This is

exciting because such interactions are difficult to detect and few if any other examples have been studied.

The authors present a careful comparison of their method to another similar approach, quantitative trait locus (QTL) analysis, and demonstrate that in situations such as the one analysed it has greater power to detect genetic factors with a certain magnitude of effect. They also test the statistical properties of their method using simulated data and permutation tests. Overall the analysis is rigorous and well motivated, and the methods explained clearly.

The main limitation of the approach is that it is difficult to see how it might be applied beyond the context of mutation accumulation experiments using recombinant inbred lines. This is because the signal it detects, and hence its power, is based on the number of extra accumulated mutations linked to (i.e. on the same chromosome as) the mutator allele. In germline mutation studies of wild populations the number of generations involved (and hence the total number of mutations) is typically small, or else the mutator allele becomes unlinked from the mutations it has caused (due to recombination), or is lost from the population altogether (due to chance or perhaps selection against its deleterious consequences).

Nevertheless, accumulation lines are a common and well established experimental approach to studying mutation processes in many organisms, so the new method could have wide application and impact on our understanding of this fundamental biological process.

The evidence presented for an epistatic interaction is convincing, and the authors suggest some plausible potential mechanisms for how this interaction might arise, involving the DNA repair machinery and based on previous studies of the proteins implicated. However as with all such findings, given the higher degree of complexity of the proposed model it needs to be treated with greater caution, perhaps until replicated in a separate dataset or demonstrated in follow-up experiments exploring the pathway itself.

Reviewer #3 (Public Review):

Sasani et al. develop and implement a new method for mutator allele discovery in the BXD mouse population. This new "IHD" method carries several notable strengths, including the ability to aggregate de novo mutations across individuals to reduce data sparsity and to combine mutation rate frequencies across multiple nucleotide contexts into a single estimate. These advantages may render the IHD method better suited to mutator discovery under certain scenarios, as compared to conventional QTL or association mapping. Overall, the theoretical premise of the IHD method is judged to be both strong and innovative, and careful simulation studies benchmark its power.

The authors then apply their method to the BXD mouse recombinant inbred mapping population. As proof-of-principle, they first successfully re-identify a known mutator locus in this population on chr4. Next, to assess possible genetic interactions involving this known mutator, Sasani et al. condition on the chr4 mutator genotype and reimplement the IHD scan. This strategy led them to identify a second locus on chr6 that interacts epistatically with the chr4 locus; mice with "D" alleles at both loci exhibit a significantly increased burden of C>A de novo mutations, even though mice with the D allele at the chr6 locus alone show no appreciable increase in the C>A mutation fraction. This exciting discovery not only adds to the catalog of known mutator alleles, but also reveals key aspects of mutator biology. Notably, this finding reinforces the hypothesis that segregating variants in genes associated with DNA repair influence germline mutation spectra. Further, Sasani et al.'s findings suggest that some mutators may lie dormant until recombined onto a permissive genetic background. This discovery could have intriguing implications for the evolution of mutators in natural populations.

Despite a high level of overall enthusiasm for this work, some weaknesses are identified in the IHD method, approach for nominating candidate genes within the newly identified chr6 locus, and the authors' conclusions.

Under simulated scenarios, the authors' new IHD method is not appreciably more powerful than conventional QTL mapping methods. While this does not diminish the rigor or novelty of the authors findings, it does temper enthusiasm for the IHD method's potential to uncover new mutators in other populations or datasets. Further, adaptation of this methodology to other datasets, including human trios or multigenerational families, will require some modification, which could present a barrier to broader community uptake. Notably, BXD mice are (mostly) inbred, justifying the authors consideration of just two genotype states at each locus, but this decision prevents out-of-the-box application to outbred populations and human genomic datasets. Lastly, some details of the IHD method are not clearly spelled out in the paper. In particular, it is unclear whether differences in BXD strain relatedness due to the breeding epoch structure are fully accounted for in permutations. The method's name - inter-haplotype distance - is also somewhat misleading, as it seems to imply that *de novo* mutations are aggregated at the scale of sub-chromosomal haplotype blocks, rather than across the whole genome.

Nominating candidates within the chr6 mutator locus requires an approach for defining a credible interval and excluding/including specific genes within that interval as candidates. Sasani et al. delimit their focal window to 5Mb on either side of the SNP with the most extreme P-value in their IHD scan. This strategy suffers from several weaknesses. First, no justification for using 10 Mb window, as opposed to, e.g., a 5 Mb window or a window size delimited by a specific threshold of P-value drop, is given, rendering the approach rather ad hoc. Second, within their focal 10Mb window, the authors prioritize genes with annotated functions in DNA repair that harbor protein coding variants between the B6 and D2 founder strains. While the logic for focusing on known DNA repair genes is sensible, this locus also houses an appreciable number of genes that are not functionally annotated, but could, conceivably, perform relevant biological roles. These genes should not be excluded outright, especially if they are expressed in the germline. Further, the vast majority of functional SNPs are non-coding, (including the likely causal variant at the chr4 mutator previously identified in the BXD population). Thus, the author's decision to focus most heavily on coding variants is not well-justified. Sasani et al. dedicate considerable speculation in the manuscript to the likely identity of the causal variant, ultimately favoring the conclusion that the causal variant is a predicted deleterious missense variant in Mbd4. However, using a 5Mb window centered on the peak IHD scan SNP, rather than a 10Mb window, Mbd4 would be excluded. Further, SNP functional prediction accuracy is modest [e.g., PMID 28511696], and exclusion of the missense variant in Ogg1 due its benign prediction is potentially premature, especially given the wealth of functional data implicating Ogg1 in C>A mutations in house mice. Finally, the DNA repair gene closest to the peak IHD SNP is Rad18, which the authors largely exclude as a candidate.

Additionally, some claims in the paper are not well-supported by the author's data. For example, in the Discussion, the authors assert that "multiple mutator alleles have spontaneously arisen during the evolutionary history of inbred laboratory mice" and that "...mutational pressure can cause mutation rates to rise in just a few generations of relaxed selection in captivity". However, these statements are undercut by data in this paper and the authors' prior publication demonstrating that a number of candidate variants are segregating in natural mouse populations. These variants almost certainly did not emerge *de novo* in laboratory colonies, but were inherited from their wild mouse ancestors. Further, the wild mouse population genomic dataset used by the authors falls far short of comprehensively sampling wild mouse diversity; variants in laboratory populations could derive from unsampled wild populations.

Finally, the implications of a discovering a mutator whose expression is potentially conditional on the genotype at a second locus are not raised in the Discussion. While not a weakness per se, this omission is perceived to be a missed opportunity to emphasize what, to this reviewer, is one of the most exciting impacts of this work. The potential background dependence of mutator expression could partially shelter it from the action of selection, allowing the allele persist in populations. This finding bears on theoretical models of mutation rate evolution and may have important implications for efforts to map additional mutator loci. It seems unfortunate to not elevate these points.